

GREATER BINGHAMTON, NY, USA

WMO: 725150

Lat: 42.207N

Lon: 75.980W

Elev: 1595

StdP: 13.87

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04725

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	3.9	-9.2	3.5	1.6	-4.6	4.6	5.7	24.7	27.3	22.3	22.8	10.0	300	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	85.1	70.0	82.3	68.6	79.8	67.4	72.8	80.8	71.2	78.5	69.8	76.7	9.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	117.9	76.4	68.8	112.2	75.1	67.4	106.9	73.8	37.3	80.8	36.0	78.5	34.7	76.7	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.6	18.5	16.7	DB	-5.7	89.5	5.6	2.7	-9.7	91.5	-13.0	93.1	-16.1	94.6	-20.1	96.5
			WB	-6.4	76.0	5.5	2.1	-10.3	77.5	-13.6	78.7	-16.7	79.9	-20.7	81.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	22.6	25.0	32.8	45.1	56.5	64.9	69.2	67.7	60.7	49.5	38.4	28.6
	DBStd	18.58	11.70	10.36	10.90	9.64	8.48	6.55	5.27	5.31	7.17	8.45	9.34	9.51
	HDD50	3485	851	700	547	206	31	0	0	0	5	115	365	665
	HDD65	7034	1314	1119	1000	602	288	82	20	33	167	482	799	1128
	CDD50	2339	1	1	12	58	233	447	595	548	326	101	15	2
	CDD65	415	0	0	0	3	26	79	150	116	38	3	0	0
	CDH74	2565	0	0	4	41	198	498	939	651	225	9	0	0
	CDH80	522	0	0	0	7	33	98	211	132	41	0	0	0
Wind	WSAvg	8.3	9.4	9.5	9.6	9.4	8.2	7.4	6.8	6.6	7.2	8.2	8.8	9.1
Precipitation	PrecAvg	37.9	2.5	2.3	2.9	3.3	3.5	3.9	3.4	3.3	3.6	3.0	3.2	2.9
	PrecMax	49.0	6.4	4.5	6.0	8.6	7.0	9.4	7.7	6.8	9.6	7.9	7.5	6.1
	PrecMin	30.3	0.7	0.5	0.7	0.7	0.8	1.0	1.5	1.4	0.6	0.3	1.0	0.9
	PrecStd	5.5	1.2	1.1	1.2	1.6	1.7	2.1	1.7	1.3	2.0	1.8	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.4	56.8	69.7	79.7	83.9	87.2	88.7	88.1	84.8	74.6	65.0	59.7
		MCWB	55.1	50.3	56.5	60.3	66.7	71.7	72.0	70.5	68.1	63.8	55.6	54.8
	2%	DB	50.2	49.4	60.5	72.0	79.5	82.7	85.0	83.6	79.9	70.1	60.3	52.1
		MCWB	46.9	42.4	49.8	56.2	63.9	68.9	70.5	69.5	67.4	60.6	53.8	48.3
	5%	DB	44.4	44.6	54.7	65.8	75.2	79.7	82.2	80.3	76.0	66.6	56.9	46.7
		MCWB	41.5	39.4	47.1	53.0	62.2	67.8	69.3	67.8	65.7	59.3	51.1	43.1
	10%	DB	38.4	39.6	49.2	60.8	71.0	76.5	79.3	77.6	72.4	63.0	53.1	41.9
		MCWB	35.9	35.4	42.5	50.3	59.7	65.7	68.0	66.9	63.8	56.8	48.5	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	52.1	57.4	62.8	70.0	73.8	75.4	74.5	72.3	67.2	58.9	55.4
		MCDB	57.1	56.4	67.0	75.1	80.8	82.5	84.4	82.7	79.5	71.5	62.1	59.1
	2%	WB	48.0	44.2	52.4	58.8	67.1	71.5	72.9	72.0	69.6	63.9	55.4	49.3
		MCDB	49.4	48.2	58.4	67.6	75.8	79.6	81.5	78.5	76.6	67.3	58.9	51.8
	5%	WB	41.7	39.9	47.8	55.5	64.3	69.8	71.4	70.3	67.7	60.5	52.3	44.0
		MCDB	44.2	43.8	53.9	63.9	71.8	76.8	78.9	76.8	73.2	65.2	55.8	46.5
	10%	WB	36.1	35.5	43.0	51.6	61.9	67.9	69.9	68.7	65.6	57.2	48.6	38.8
		MCDB	38.4	39.0	49.0	58.1	69.1	74.1	76.7	75.0	70.4	62.2	52.6	41.7
Mean Daily Temperature Range	5% DB	MDBR	12.8	13.8	15.4	17.8	18.1	17.0	17.0	16.7	17.1	15.0	13.0	11.0
		MCDBR	16.9	19.1	22.1	25.1	22.5	20.1	19.3	19.3	20.3	19.1	17.9	15.7
		MCWBR	15.9	15.2	14.5	14.0	11.7	10.1	8.8	8.7	9.7	11.5	13.1	13.7
	5% WB	MCDBR	16.5	17.6	20.9	22.8	20.2	17.7	16.9	16.4	17.3	16.6	16.4	15.3
		MCWBR	16.1	15.3	15.5	15.1	11.9	10.1	9.0	8.6	9.4	11.6	13.9	14.7
Clear-Sky Solar Irradiance	taub	0.289	0.306	0.328	0.350	0.381	0.431	0.437	0.420	0.356	0.343	0.302	0.284	
	taud	2.385	2.339	2.390	2.391	2.334	2.215	2.207	2.245	2.411	2.452	2.522	2.479	
	Ebn at Noon	272	285	291	292	283	268	265	266	278	268	267	266	
	Edh at Noon	25	31	34	36	39	44	44	42	33	28	22	21	
All-Sky Solar Radiation	RadAvg	470	747	1091	1396	1649	1752	1811	1603	1291	821	539	374	
	RadStd	38	101	97	162	166	164	134	93	138	95	67	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (28 neighbors)	+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-187	+189	+83

Nomenclature: See separate page